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 Studierichting: Informatica
 Oefeningenreeks 5J nr. 2.1

$$\begin{aligned}
 & \begin{cases} x(t) = a(t - \sin t) \\ y(t) = a(1 - \cos t) \end{cases} \\
 L &= \int_0^{2\pi} \sqrt{((a(t - \sin t))')^2 + ((a(1 - \cos t))')^2} dt \\
 &= \int \sqrt{((a(t - \sin t))')^2 + ((a(1 - \cos t))')^2} dt \\
 &= \int \sqrt{(a - a \sin t)^2 + (a - a \cos t)^2} dt \\
 &= \int \sqrt{(a - a \cos t)^2 + (a \sin t)^2} dt \\
 &= a \int \sqrt{(1 - \cos t)^2 + (\sin t)^2} dt \\
 &= a \int \sqrt{1 - 2 \cos t + \cos^2 t + \sin^2 t} dt \\
 &= a \int \sqrt{2 - 2 \cos t} dt \\
 &= a\sqrt{2} \int \sqrt{1 - \cos t} dt \\
 &= a\sqrt{2} \int \sqrt{1 - 1 + 2 \sin^2 \frac{t}{2}} dt \\
 &= a\sqrt{2} \int \sqrt{2} \sin \frac{t}{2} dt \\
 &= 2a \int \sin \frac{t}{2} dt \\
 &= 4a \left(-\cos \left(\frac{t}{2} \right) \right) + c \\
 \Rightarrow L &= 4a \left[-\cos \left(\frac{t}{2} \right) \right]_0^{2\pi} = 4a(-\cos \pi + \cos 0) = 8a
 \end{aligned}$$

Grafiek (voor $a = 1$):

References

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